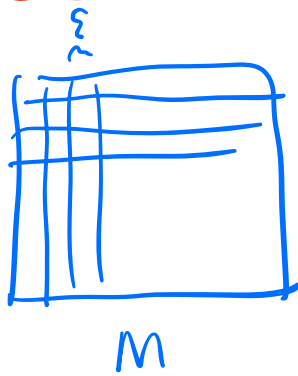
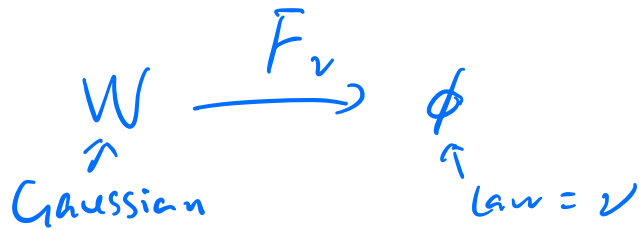


Lecture 3 Langevin dynamic



$\Lambda_{\xi, M}$

Stochastic quantization



General fact:

$$dX_t = U'(X_t)dt + dW_t \quad W_t \in \mathbb{R}^D \quad D = |\Lambda_{\xi, M}|$$

leaves $\frac{1}{Z} e^{2U(y)} d^D y$ invariant. $W_t(x): 1d \text{ BM}$
 $\forall x \in \Lambda_{\xi, M}$

Then take $\phi = X(t=0)$

EX: ① GFF on $\Lambda_{\xi, M}$.

$$\frac{1}{Z} e^{-\frac{1}{2} \sum_{x \in \Lambda_{\xi, M}} X^{(x)} (m^2 - \Delta_\xi) X^{(x)}} \prod_x dX^{(x)}$$

$$X \sim N(0, (m^2 - \Delta_\xi)^{-1}) =: \mu^{\xi, M}$$

$$dX_t = (\Delta_\xi - m^2) X_t dt + dW_t$$

$$\textcircled{2} \frac{1}{Z} e^{-\varepsilon^d \sum_{x \in \Lambda_{\xi, M}} V(\varphi(x))} \mu^{\xi, M}(d\varphi)$$

$$d\varphi_t = (\Delta_\xi - m^2) \varphi - V'(\varphi(t, x)) + dW(t, x)$$

1st Pf: Kolmogorov forward eq (Fokker-Planck)

$$\partial_t P = L^* P$$

L : generator

$$= \frac{1}{2} \operatorname{div} \nabla P - \operatorname{div} (U'(y) P)$$

where p is prob density on $\mathbb{R}^d$

$$\frac{1}{2} \nabla p - U'(y) p = 0 \quad \text{if } p = e^{2U}$$

2nd pf : Girsanov (prove stronger result : reversibility)

Claim: $\int E^y f(X_t) g(y) e^{2U(y)} dy = \int f(y) E^y g(X_t) e^{2U(y)} dy$

Rmk: ① essentially Dirichlet form "symmetric"

② set $g \equiv 1$. $\underbrace{\int E^y f(X_t) e^{2U(y)} dy}_{= E f(X_t)} = \underbrace{\int f(y) e^{2U(y)} dy}_{= E f(X_0)}$

$X_0 \sim e^{2U}$

This means invariance.

③ meaning of "reversible": pretend $f = 1_z$, $g = 1_{z_2}$

then "claim" means $P(X_0 = z_2, X_t = z_1) = P(X_0 = z_1, X_t = z_2)$

pf: $E^y f(X_t) \stackrel{\text{Girsanov}}{=} E^y [f(W_t) e^{\int_0^t \nabla U(W_s) dW_s - \frac{1}{2} \int_0^t |\nabla U(W_s)|^2 ds}]$

$\uparrow_{\text{BM}}$

(By Itô. $dU(W_t) = \nabla U(W_t) dW + \frac{1}{2} \Delta U(W_t) dt$)

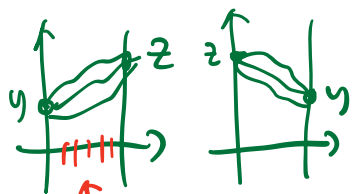
$= E^y [f(W_t) e^{U(W_t) - U(W_0) - \frac{1}{2} \int_0^t |\nabla U(W_s)|^2 + \Delta U(W_s) ds}]$

"Condition on $W_t = z$ "

$= \int dz P_t(z-y) f(z) e^{U(z) - U(y)} E^y [e^{-\frac{1}{2} \int_0^t |\nabla U(W_s)|^2 + \Delta U(W_s) ds} | W_t = z]$

$= \int dz P_z(y-z) f(z) e^{U(z) - U(y)} E^z [e^{-\frac{1}{2} \int_0^t |\nabla U(W_s)|^2 + \Delta U(W_s) ds} | W_t = y]$

Here we used $\mathbb{E}^y [e^{\int_0^t H(W_s) ds} | W_t = z] = \mathbb{E}^z [e^{\int_0^t H(W_s) ds} | W_t = y]$
 BM reversal property



evenly divide to $n+1$ pieces
 $s_1 < s_2 < \dots < s_n$

$$\int e^{\frac{1}{n} \sum_{i=1}^n H(y_i)} P_{s_1}(y_1, y) \dots P_{t-s_n}(z, y_n) dy_1, \dots, dy_n$$

$y \rightarrow y_1 \rightarrow y_2 \rightarrow \dots \rightarrow y_n \rightarrow z$
 $s_1 = s_2 - s_1 = \dots = t - s_n = \frac{t}{n+1}$

LHS of "claim"

$$= \iint P_z(y, z) f(z) e^{u(z)+u(y)} \mathbb{E}^z [e^{-\frac{1}{t} \int_0^t |\nabla u(W_s)|^2 + \delta u(W_s) ds} | W_t = y] \cdot g(y) dz dy$$

exchange $z \leftrightarrow y$ = RHS of "claim"

Rmk: Our t here is fictitious time, nothing to do with phys time (like "time" of Minkowski on Monday). "OU" on Monday is over space $\mathbb{R}^d$.

We will focus on Φ^4 model

$$V(\varphi) = \frac{\beta}{2} \varphi(x)^2 + \frac{\lambda}{4} \varphi(x)^4 \quad \begin{matrix} \lambda > 0 \\ \beta \in \mathbb{R} \end{matrix}$$

Rmk: ① Novikov condition etc (omit)

② V is not globally Lipschitz. (standard well-posedness doesn't apply)

$$dX = (\Delta_\varepsilon - m^2)X - \lambda X^3 - \beta X dt + dW$$

rule out finite time blow up?

Below we give two arguments.

$$\begin{aligned}
 \textcircled{1} \quad d \sum_{x \in \Lambda_{\varepsilon, m}} X_t(x)^2 &\stackrel{\text{Itô}}{=} \sum_x X_t(x) dX_t(x) + \underbrace{\sum_x dt}_{\text{Itô term}} \\
 &= \underbrace{\sum_x X_t \left((\Delta_\varepsilon - m^2) X_t - \lambda X_t^3 \right)}_{\text{negative}} dt + \underbrace{(\quad) dw}_{\text{martingale}} \\
 &\quad + |\Lambda_{\varepsilon, m}| dt - \beta \sum_x X_t(x)^2 dt
 \end{aligned}$$

$$d \left(\mathbb{E} \sum_x X_t(x)^2 \right) \leq |\beta| \left(\mathbb{E} \sum_x X_t(x)^2 \right) dt + |\Lambda_{\varepsilon, m}| dt$$

$$\left(\begin{aligned}
 \text{Gronwall: } \underline{u(t)} &\leq \alpha(t) + \int_0^t \beta(s) \underline{u(s)} ds \\
 \Rightarrow u(t) &\leq \alpha(t) + \int_0^t \alpha(s) \beta(s) e^{\int_s^t \beta(r) dr} ds
 \end{aligned} \right)$$

$$\text{Thus } \underbrace{\mathbb{E} \sum_x X_t(x)^2}_{\text{so, a.s. finite}} < \infty \quad \forall t.$$

② Second argument more robust for future studies.

Do not use Itô.

"DaPrato-Debussche" argument:

$$X = Y + Z \quad \text{Let } A = \Delta_\varepsilon - m^2.$$

$$\begin{cases} dY = AY + dw \\ \frac{dZ}{dt} = AZ - V'(Y+Z) \end{cases}$$

↑ random

$$\left(\begin{aligned}
 &\text{no Itô} \\
 \text{eg. } dZ &= B dt \\
 \left(\frac{1}{2} d(Z^2) \right) &= Z \cdot B \cdot dt
 \end{aligned} \right)$$

$$\text{multiply } Z \text{ on both sides: LHS: } \frac{dZ}{dt} \cdot Z \stackrel{\uparrow}{=} \frac{1}{2} \frac{d}{dt} (Z^2)$$

$$\frac{1}{2} \frac{d}{dt} \sum_x Z_x(x)^2 + \underbrace{\sum_x Z_x (m^2 - \Delta_\varepsilon) Z_x + \lambda \sum_x Z_x^4}_{=: G(Z_\varepsilon) \geq 0}$$

$$= \sum_x \lambda (Y^3 Z + 3 Y^2 Z^2 + 3 Y Z^3) + \beta \sum_x (Z^2 + 2 Y Z + Z^2)$$

$$\text{Claim} \leq C_\delta \cdot \|Y_t\|_{\ell^4(\Lambda_{\varepsilon, m})}^4 + \delta \cdot G(Z_\varepsilon)$$

for $\delta > 0$ small and C_δ dep on δ .

Indeed, recall Holder ineq $\sum_x f_x g_x \leq \left(\sum_x f_x^p\right)^{1/p} \left(\sum_x g_x^q\right)^{1/q}$
 $\frac{1}{p} + \frac{1}{q} = 1$

$$\text{Young ineq } ab \leq \frac{a^p}{p} + \frac{b^q}{q} \longrightarrow \leq \frac{1}{p} \sum_x f_x^p + \frac{1}{q} \sum_x g_x^q$$

Then for example:

$$\sum_x Y(x)^3 Z(x) \leq C_\delta \sum_x Y(x)^4 + \delta \cdot \sum_x Z(x)^4 \quad p = \frac{4}{3} \quad q = 4$$

Therefore

$$\|Z_t\|_{\ell^2}^2 + \frac{1}{2} \int_0^t G(Z_s) ds \leq \|Z_0\|_{\ell^2}^2 + C \int_0^t \|Y_s\|_{\ell^4}^4 ds$$

OU, no blow up
 $\downarrow$

$$Z_t \text{ no blow up. } \|Z_t\|_{\ell^2}^2 \leq \|Z_0\|_{\ell^2}^2 + C \int_0^t \|Y_s\|_{\ell^4}^4 ds$$

• Maybe we should mention uniqueness of solution here

Two solutions Z_1, Z_2 from same i.c. driven by Y . Let $Z = Z_1 - Z_2$

Z satisfies eqn of form $dZ = W \cdot Z dt, Z(0) = 0$

where $W = \text{Poly}(Z_1, Z_2, Y)$ Gronwall; $Z(t) = Z(0) e^{\int_0^t W_s ds} = 0$

Questions for rest of the school:

Can we use this dynamic to

- 1) say sth on infinite volume limit of measure as $M \rightarrow \infty$? "infrared"
- 2) say sth on Continuum limit of measure as $\varepsilon \rightarrow 0$?
- 3) prove other properties (e.g. "axioms") of the measure? "ultrafilter"

If time permits, briefly discuss other models Liouville, SG,
linear σ -model. Abelian gauge etc.

and exemplify their Langerin dynamics.

(If no time, leave this as exercise)

If there's more time, briefly discuss elliptic SQ,
canonical SQ, Variational rep etc.